

Course summary

This is a rough summary of what we have covered during our time together. The final exam will emphasize more of the material after the midterm, although material from the first four lectures will appear as well. (Look at the past final examination papers to see a good representation of the emphasis.)

Main topics:

1. Supply and demand. In particular, what causes these curves to shift, and how to use analysis to consider such things as government policies to affect supply and demand. Aggregation of supply and demand. Differences between long-run and short-run curves.
2. Elasticities. How to calculate and interpret arc and point elasticity formulae for own-price, cross-price, income and supply elasticities.
3. Indifference curves, utility functions and the concept of MRS.
4. Budget constraints, additional constraints (e.g., time constraints, government constraints, etc.)
5. Optimal choice by consumers. Two rules: budget line and the tangency rule. Various interpretations of the tangency rule (e.g. graphical analysis, capital budgeting, arbitrage).
6. Consumer surplus. What it means and how to calculate it.
7. Basic uncertainty theory. Idea of expected-utility theory, that preferences can be represented by $E[u(x)]$. Relationships between risk preferences (e.g., risk aversion) and marginal utility of income (e.g., the shape of $u(x)$). Definitions of risk premium and certainty equivalent.
8. Concept of adverse selection.
9. Opportunity costs and economic profit.
10. Short-run and long-run costs. The basic mechanics of short-run marginal and average cost; long-run marginal and average cost. The relationship between short-run accounting cost and short-run opportunity cost.
11. Production, isocosts, isoquants, MRTS. Two rules for cost minimization: no waste, $q = f(L, K)$, and “tangency”, $MRTS_{L,K} = \frac{w}{r}$.
12. Two rules for profit maximization: marginal output rule and shutdown/operation rule. How to implement these rules graphically, mathematically, etc.
13. Application of profit maximizing rules to competitive firms. Derivation of a firm’s short-run and long-run supply curves. Derivation of market short-run and long-run supply curves from firms’ supply curves. Variations on long-run supply: constant-cost industry, increasing-cost industry, firms with different costs.
14. Characterization of firm profits from firm supply curves and industry profits from industry supply curves. Source of short-run and long-run profits (e.g., rents, competitive advantages in cost, scarce and valuable resources, Ricardian rents, etc.)

15. Applications of competitive model to (a) quantity controls (e.g., taxi licenses, steel quotas, etc.), (b) unit taxes, (c) unit subsidies, (d) price controls (e.g. rent control, minimum wage laws, etc.), (e) tariffs, etc. Problem of externalities and solutions.
16. Basic monopoly theory. Nature of marginal revenue curves for firms with market power. Applying the two rules of profit maximization. Characterizing profits in a graph. Simple numerical examples, including the taxicab example in which one firm behaves as a monopolist and the other firms dump their full capacities on the market.
17. Applications of monopoly: Optimal markups using elasticities, multi-product pricing [not on final exam], taxes and subsidies, pricing with short run and long run demand elasticities [not on final exam], vertical relationships and double marginalization [not on final exam], optimal product design and the choice of quality [not on final exam], and optimal output by a monopoly/dominant firm facing a competitive fringe of price taking firms.
18. Oligopoly. The notion of an oligopoly as a game and the concept of Nash equilibrium. Output (i.e., Cournot) games. [Note: Price (i.e., Bertrand) games will not be on final exam.]
19. Price discrimination theory. Perfect price discrimination (1st degree).
20. Direct price discrimination (e.g., 3rd degree). Solving the monopoly problem multiple times. Key observations include: if MC is the same for producing across segment, then the MR's for each segment will also be the same; more generally, the difference in MC between segments will equal the difference in MR between segments. Segments may be linked via costs.
21. Indirect (second-degree) price discrimination. (a) Nonlinear-pricing/quantity-discount model (see Dolan and Simon: "Nonlinear Pricing", ch 7). (b) Simple airline pricing model of with high and low quality. Relationship between quality-distortion model and nonlinear-pricing/quantity-discount model. (c) 2-part tariffs [not on final exam].

I think the above 21 items are the main topics we covered. Of course, if there is something in your notes that I discussed in class that is not on this list above, it is possible that it will be tested on the exam.

Key micro concepts to remember years from now:

- respect markets; supply and demand are powerful forces; know your S&D;
- when price ratios are unequal, something's suboptimal and/or someone can make money;
- sunk costs are irrelevant – ignore them for decision making; make decisions based upon the appropriate marginal costs and benefits;
- making economic profit in a competitive market requires a competitive advantage (quality, cost, info, etc.); focus on scarce and valuable resources;
- market power is all about MR; profit can be improved by selling to that last customer segment if this can be done without sacrificing profit on higher-paying customers.